## Example 2.2

A sample of 10 students from a school was selected. Their scores in a particular subject are 72, 82, 96, 85, 84, 75, 76, 93, 94 and 93. Can we support the claim that the class average scores is 90?

***Solution:***

**Step 1 : Hypotheses**

**Null Hypothesis** $H_0$: $\mu = 90$

*i.e.,* the class average scores is not significantly different from 90.

**Alternative Hypothesis** $H_1 : \mu \neq 90$

*i.e.,* the class means scores is significantly different from 90.

It is a two-sided alternative hypothesis.

**Step 2 : Data**

The given sample information are

Size of the sample $(n) = 10$. Hence, it is a small sample.

**Step 3 : Level of significance**

$\alpha = 5\%$

**Step 4 : Test statistic**

The test statistic under $H_0$ is $T = \dfrac{\overline{X} - \mu_0}{S/\sqrt{n}}$

Since $n$ is small, the sampling distribution of $T$ is the $t$ - distribution with $(n-1)$ degrees of freedom.

**Step 5 : Calculation of test statistic**

The value of $T$ for the given sample information is calculated from $t_0 = \dfrac{\overline{x} - \mu_0}{s/\sqrt{n}}$ as under:

| $x_i$ | $u_i = x_i - A$; $(A = 85)$ | $u_i^2$ |
|---|---|---|
| 72 | –13 | 169 |
| 82 | –3 | 9 |
| 96 | 11 | 121 |
| 85 | 0 | 0 |
| 84 | –1 | 1 |
| 75 | –10 | 100 |
| 76 | –9 | 81 |
| 93 | 8 | 64 |
| 94 | 9 | 81 |
| 93 | 8 | 64 |
| | $\sum_{i=1}^{10} u_i = 0$ | $\sum_{i=1}^{10} u_i^2 = 690$ |



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